

Competitive Dynamic Pricing · Nash Equilibria

WEEK 2 OF 8

Multi-Agent RL · Oligopoly Simulation · Collusion Detection

PHASE 1 — FOUNDATIONS

Week 2: Market Environment Design — Build the custom OpenAI Gym environment this week. A buggy environment silently corrupts every result that follows. Unit-test the analytical Nash price output before writing a single agent.

Priority key:

**MUST WATCH /
READ**

RECOMMENDED

OPTIONAL

■ = estimated time | pp. = page range

■ YouTube & Interactive Videos

■ What is Reinforcement Learning? (Simple Explanation)

Arxiv Insights

youtube.com/watch?v=JgvyzlkqxF0

Cleanest conceptual intro to agent-environment interaction and reward signals. Grounds students in the mental model before touching gym code.

MUST WATCH

■ 10 min

■ OpenAI Gym Tutorial — Getting Started

sentdex

youtube.com/watch?v=8MC3y7ASoPs

Quickest practical intro to the gym interface — observation/action spaces, step(), and reward loops. The exact template for the pricing environment.

MUST WATCH

■ 18 min

■ Building Custom OpenAI Gym Environments

Nicholas Renotte

youtube.com/watch?v=bD6V3rcr_54

Step-by-step tutorial on subclassing gym.Env with full code. Students should code along then adapt the structure to the Bertrand pricing market.

MUST WATCH

■ 32 min

■ Linear Demand Functions — Economics Tutorial

EconplusDal

youtube.com/watch?v=XSqKKJxGGhM

Students must understand $Q = a - bP$ before coding it. Covers demand curves, elasticity, and inverse demand — all needed for the profit function.

MUST WATCH

■ 8 min

■ Reinforcement Learning in 3 Hours | Full Course

Nicholas Renotte

youtube.com/watch?v=Mut_u40Sqz4

Watch the first 45 min for environment setup context. A comprehensive practical RL course — return to later sections in Weeks 4 and 5.

RECOMMENDED

■ 45 min (first 45 min only)

■ Python OOP in 90 Minutes — Classes and Inheritance

Tech With Tim

youtube.com/watch?v=JeznW_7DIB0

If students are shaky on OOP, this is the fastest fix. The market environment relies on clean class design — weak OOP creates bugs throughout the project.

OPTIONAL

■ 90 min

■ Papers, Books & Blogs

■ Gymnasium: Creating Custom Environments

Farama Foundation — Official Docs

gymnasium.farama.org → Environment Creation

The official step-by-step guide. Read 'Make your own custom environment' and implement alongside the tutorial.

MUST READ

■ ~30 min

Web —

gymnasium.farama.org

■ Reinforcement Learning: An Introduction — Ch. 1 & 2

Sutton & Barto (Free PDF)

incompleteideas.net/book/RLbook2020.pdf

Ch. 1 frames RL as agent-environment interaction. Ch. 2 covers bandit problems as a warm-up for Q-learning in Week 4.

MUST READ

■ ~40 min

pp. 1–37

■

CartPole-v1 Source Code

Farama Foundation — GitHub

github.com/Farama-Foundation/Gymnasium → [cartpole.py](#)

Read this ~100-line env to see `step()` / `reset()` structure before writing your own. The cleanest available template.

MUST READ

■ ~15 min

~100 lines

➡

Build a Custom OpenAI Gym Environment from Scratch

Towards Data Science

towardsdatascience.com → [search 'Custom Gym Environment'](#)

Practical walkthrough with full working code. Adapt the environment class to the Bertrand pricing market step by step.

RECOMMENDED

■ ~20 min

Web article

- Common Pitfalls to Watch — Week 2
- Discretising the price action space too coarsely (e.g. 5 levels) — use 20–50 price levels minimum.

■ Hard-coding demand parameters — make a , b , and marginal cost configurable from day one.

■ Forgetting to normalise observations — design the env with normalisation in mind before agents are added.

■ Videos (must)	■ Videos (rec.)	■ Reading (must)	■ Total (must only)	■ Total (all items)
68 min	45 min	85 min	~2.5 hr	~5 hr